Huawei Access Node HEB-ACC-75 Hebbal Edge QoS/Congestion Event Mitigation SOP (ALARM_5636)

Node: HEB-ACC-75 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_5636

Description

This Standard Operating Procedure (SOP) addresses the mitigation of QoS/Congestion Events (Microbursts, interface tail drops, QoS queue starvation) on Huawei Access Node HEB-ACC-75 at the Hebbal Edge site. The underlying cause of this failure domain lies in the complex interactions between the Huawei Access Node's hardware and software components, specifically the Queue Management and Scheduling (QMS) and Enhanced Quality of Service (QoS) modules. QMS is responsible for managing the queue of cells/packets in the node, while QoS ensures that the packet forwarding order is based on the priority assigned to each packet. However, under high traffic conditions or when the network is congested, microbursts or tail drops can occur due to the QMS's inability to handle the packet queue effectively. Additionally, QoS queue starvation may occur when high-priority packets are consistently dropped or delayed, leading to a decrease in overall network performance. This SOP aims to identify the root cause of the QoS/Congestion Event, perform necessary diagnostic tests, and implement corrective actions to prevent future occurrences.

Pre-requisite Checks

- Ensure the Hebbal Edge site is accessible and remote hands are dispatched to verify the physical environment.

- Verify the status of redundant links to ensure proper backup in case of a link failure.

- Run a basic health check on the Huawei Access Node HEB-ACC-75 using the command 'display current-configuration | include node-id' to confirm the node's identity and configuration.

Potential Root Causes

Cause	Probability
Inadequate Queue Management and Scheduling (QMS) configuration	High
Insufficient Enhanced Quality of Service (QoS) configuration	Medium
Hardware-related issues with the Access Node's interface or backplane	Low

Diagnostic Commands & Expected Outputs

```
Command: display queue-statistics, Expected Output: Queue statistics for all queues in the node, Description: This command displays the current queue statistics for all queues in the node, including the number of cells/packets in each queue, the average queue length, and the number of cells/packets dropped due to queue overflow.
```

```
Command: display qos statistics, Expected Output: QoS statistics for all queues in the node, Description: This command displays the current QoS statistics for all queues in the node, including the number of high-priority packets dropped or delayed, and the average QoS queue length.
```

Resolution Steps

Step 1: Step: Verify QMS Configuration, Command: display qms configuration, Description: Check the QMS configuration to ensure that it is properly configured to handle the current traffic load. If necessary, adjust the QMS parameters to optimize queue management.

Step 2: Step: Verify QoS Configuration, Command: display qos configuration, Description: Check the QoS configuration to ensure that it is properly configured to handle high-priority packets. If necessary, adjust the QoS parameters to optimize packet forwarding order.

Step 3: Step: Clear Queue Statistics, Command: clear queue-statistics, Description: Clear the queue statistics to reset the counters and obtain accurate measurements after the corrective actions are taken.

Post-Check Verification

```
Command: display queue-statistics, Expected Output: Queue statistics for all queues in the node, Description: Verify that the queue statistics have returned to normal after the corrective actions were taken.

Command: display qos statistics, Expected Output: QoS statistics for all queues in the node, Description: Verify that the QoS statistics have returned to normal after the corrective actions were taken.
```

Roll-back Procedures

- Step: Restore Original QMS Configuration, Command: apply qms configuration , Description: Restore the original QMS configuration if the corrective actions caused any issues.

- Step: Restore Original QoS Configuration, Command: apply qos configuration , Description: Restore the original QoS configuration if the corrective actions caused any issues.